When Volatility Targeting Crowds: Quantifying the Tipping Point via Agent-Based Simulation*

Yi-Hao Lai† VolPred Research System

April 2026

Abstract

Volatility targeting (VT) strategies—which scale equity exposure inversely to conditional volatility—are increasingly adopted by institutional investors, yet the systemic consequences of widespread adoption remain unquantified. We develop an agent-based model (ABM) with a Kyle (1985) market maker, endogenous VIX dynamics, and 1,000 heterogeneous agents to quantify crowding effects as VT adoption rises from 0% to 100%. A fixed pool of 200 noise traders provides constant background liquidity, ensuring that performance degradation reflects crowding rather than liquidity evaporation. Across 500 Monte Carlo simulations per adoption level (500 × 2,520 days each), we identify a nonlinear tipping point: VT performance remains stable up to 30% adoption (Sharpe 0.47), degrades meaningfully at 50% (Sharpe 0.34, a 28% decline), and collapses at 70% (Sharpe 0.08, an 82% decline). The model encodes a Brunnermeier–Pedersen (2009) positive feedback structure—correlated VT selling amplifies price declines, raises realized volatility, and triggers further selling—and we quantify its threshold behavior under controlled conditions. At 100% adoption, VT Sharpe is −0.27 and market kurtosis reaches 61. Sensitivity analysis across 27 parameter combinations confirms that the tipping point is conditionally stable: it shifts primarily with market

*We thank the VolPred Research System for computational support and OpenAI Codex for adversarial code review. Simulation code and replication data available upon request.

†Department of Finance, Da-Yeh University. Corresponding author. Email: yhlai@mail.dyu.edu.tw

impact (λ), while VIX dynamics (γ, κ) have limited influence. A design validation comparing fixed versus scaled liquidity shows that approximately half the degradation observed in a naïve design is attributable to liquidity evaporation rather than crowding per se. Current real-world VT adoption is estimated below 5%, suggesting the strategy remains safe for individual adopters but highlighting a trajectory that warrants monitoring.

Keywords: volatility targeting, crowding, agent-based model, tipping point, systemic risk, positive feedback

JEL Classification: G11, G12, G17, G18

1 Introduction

Volatility targeting (VT) has become a ubiquitous tool in dynamic portfolio management. The canonical framework, formalized by Moreira and Muir [2017], prescribes scaling equity exposure as $w_t = \sigma^*/\hat{\sigma}_t$, where σ^* is the target volatility and $\hat{\sigma}_t$ is a conditional volatility estimate. Harvey et al. [2018] demonstrate that such strategies meaningfully reduce tail risk across asset classes, and the approach has been widely adopted by pension funds, insurance companies, and target-date funds. Industry estimates suggest that short-volatility and volatility-sensitive strategies collectively manage over USD 2 trillion in assets [**?**].

Despite individual-level benefits, VT strategies embed a fundamentally procyclical mechanism: when volatility rises, all VT investors simultaneously reduce equity exposure, creating correlated selling pressure. The crowding risk literature warns of such procyclical dynamics—Baltas [2019] document crowding effects in alternative risk premia, and the European Central Bank [ECB, European Central Bank, 2020] explicitly flagged VT procyclicality as a source of market fragility—but no study provides a quantitative threshold for VT crowding. Several commentators have drawn parallels to the portfolio insurance strategies implicated in the 1987 crash [Gennotte and Leland, 1990]. Bookstaber et al. [2014] use agent-based modeling to study financial system fragility but do not specifically examine VT crowding.

A critical gap remains: *at what adoption level does VT crowding become destabilizing?* Existing analyses are either qualitative [European Central Bank, 2020], focused on other strategy types [Baltas, 2019], or lack a quantitative threshold. This matters because VT adoption is growing—from niche quantitative strategies a decade ago to mainstream institutional mandates today—yet no framework exists to assess whether or how the strategy might undermine itself.

We address this gap with an agent-based simulation that embeds VT agents within an endogenous market microstructure. Our contribution is fourfold. First, we identify a quantitative tipping point: VT strategy performance remains stable below 30% adoption but degrades nonlinearly beyond 50%, with collapse at 70%. A design validation confirms that this threshold is a genuine crowding effect, not an artifact of liquidity evaporation. Second, we do not claim to *discover* the positive feedback mechanism—which is encoded in the model structure—but rather *quantify* its threshold behavior under controlled conditions, establishing a direct link to the Brunnermeier and Pedersen [2009] liquidity spiral framework. Third, we document the market-level consequences of excessive VT adoption, including volatility amplification, kurtosis explosion, and increased flash crash frequency. Fourth, a sensitivity analysis across 27 parameter combinations shows that the tipping point shifts primarily with market impact (λ), while VIX dynamics are secondary—providing conditional robustness for the threshold estimate. These results are preliminary and model-dependent, but they provide a first quantitative benchmark for assessing VT crowding risk.

2 Model

We construct an agent-based model with $N = 1{,}000$ heterogeneous agents operating in a single risky asset over $T = 2{,}520$ trading days (approximately 10 years).

2.1 Agent Types

Agents are partitioned into three classes:

1. **Buy-and-Hold (BH)** agents maintain a constant equity weight of $w^{\text{BH}} = 1.0$.
2. **12/VIX Volatility Targeting (VT)** agents set their equity weight as $w_t^{\text{VT}} = \min(12/\text{VIX}_{t-1},\ 1.5)$, using the lagged VIX to avoid lookahead bias. This rule is a widely-used practitioner heuristic [Perchet et al., 2015].
3. **Noise traders** adjust their weights by a random increment $\Delta w \sim N(0, 0.02)$, clipped to $[0, 1.5]$, providing background liquidity.

The key experimental variable is the VT adoption fraction ϕ, tested at seven levels: 0%, 10%, 20%, 30%, 50%, 70%, and 100%. As ϕ increases, BH agents are replaced by VT agents while noise traders remain fixed at $N_{\text{noise}} = 200$. This design isolates the crowding mechanism from liquidity effects: because the noise-trader population is constant across all adoption levels, any performance degradation is attributable to correlated VT trading rather than to the mechanical reduction of background liquidity providers. Section 3.6 reports a design validation confirming that this distinction is quantitatively important.

2.2 Market Microstructure

Price formation follows a simplified Kyle [1985] model:

$$r_t = \mu + \lambda \cdot \frac{\mathrm{NOF}_t}{N} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_f), \tag{1}$$

where $\mu = 0.08/252$ is the daily drift, $\lambda = 0.005$ is the Kyle lambda (price impact per unit of normalized order flow), NOF_t is the aggregate net order flow from all agents, and $\sigma_f = 0.16/\sqrt{252}$ is the fundamental volatility. The VIX evolves endogenously:

$$\mathrm{VIX}_t = \max\Big(0,\ \mathrm{VIX}_{t-1} + \kappa\big(\bar{V} + \gamma \cdot \Delta\sigma_t^{\mathrm{real}} - \mathrm{VIX}_{t-1}\big) + \eta_t\Big), \tag{2}$$

where $\kappa = 0.03$ is the mean-reversion speed, $\bar{V} = 18$ is the long-run mean, $\gamma = 200$ amplifies deviations of 20-day realized volatility from 16%, and $\eta_t \sim N(0, 0.3)$. The VIX is bounded to $[9, 80]$.

2.3 Feedback Structure

The model is designed to embed a positive feedback loop: when VT agents sell (because VIX rises), the negative order flow depresses prices via Eq. (1), which increases realized volatility, which raises VIX via Eq. (2), which triggers further VT selling. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] loss spiral applied to the VT context; it is a modeling assumption, not a simulation output. The strength of this feedback loop is proportional to ϕ, and the simulation's purpose is to quantify the adoption threshold at which the loop becomes destabilizing.

2.4 Simulation Design

For each of the 7 adoption levels, we run $M = 500$ independent Monte Carlo simulations with different random seeds, producing a total of 3,500 simulations $\times$ 2,520 days = 8.82 million agent-day observations. Bootstrap 95% confidence intervals (2,000 replications) are reported for all key metrics. For each simulation we compute: annualized market volatility, VT strategy Sharpe ratio, maximum drawdown, excess kurtosis, skewness, flash crash frequency (daily returns exceeding 3σ), and VIX dynamics. We additionally run 200 simulations per cell across a one-at-a-time (OAT) sensitivity analysis varying λ, γ, and κ each at $\pm 50\%$ of baseline (9 parameter configurations) at three adoption levels (10%, 30%, 50%) to test threshold robustness.

3 Results

3.1 Strategy Performance Degradation

Table 1 presents the core results under fixed background liquidity ($N_{\text{noise}} = 200$). VT strategy performance exhibits a clear nonlinear pattern. At low adoption ($\phi = 10$–30%), the VT Sharpe ratio remains near 0.47–0.50, statistically indistinguishable from the uncrowded baseline. The critical transition occurs between 50% and 70%: the Sharpe ratio falls to 0.34 at 50% (a 28% decline relative to 10%) and collapses to 0.08 at 70% (an 82% decline). At 100%, VT generates substantially negative risk-adjusted returns (-0.27).

Table 1: VT Strategy and Market Outcomes by Adoption Level

	Strategy Performance				**Market Stability**			
ϕ	Sharpe	95% CI	ΔSharpe	Ret. (%)	ΔVol (%)	Kurt.	Kurt. CI	Flash/yr
0%	—	—	—	—	baseline	−0.00	[−0.01, 0.01]	0.32
10%	0.47	[0.44, 0.50]	—	4.69	−0.1	−0.01	[−0.02, −0.00]	0.30
20%	0.50	[0.47, 0.52]	+6%	4.98	+0.8	0.00	[−0.01, 0.01]	0.31
30%	0.47	[0.44, 0.50]	−0%	4.74	+3.8	−0.00	[−0.01, 0.00]	0.31
50%	0.34	[0.31, 0.36]	−28%	3.46	+18.8	0.06	[0.05, 0.07]	0.40
70%	0.08	[0.07, 0.10]	−82%	0.90	+42.7	1.41	[1.28, 1.55]	1.09
100%	−0.27	[−0.28, −0.26]	−157%	−3.82	+119.1	61.4	[59.2, 63.4]	1.20[a]

Results are means across 500 Monte Carlo simulations × 2,520 days each (N = 1,000 agents; $N_{\text{noise}} = 200$ fixed). 95% CIs from 2,000 bootstrap replications. Sharpe ratio and return are for the VT portfolio. ΔSharpe is relative to the 10% baseline. ΔVol is the percentage change in annualized market volatility relative to the 0% (no VT) scenario. Flash crash frequency counts daily returns $> 3\sigma$. Kurtosis is excess (Fisher); bootstrap CIs reported for all rows.

[a] At 100% adoption, the moderate flash crash frequency relative to 70% reflects a measurement artifact: the standard deviation used to define the 3σ threshold is itself severely inflated (annualized vol of 35%), masking extreme events.

The nonlinearity is striking. Performance is essentially flat from 10% to 30%, then drops sharply: a linear extrapolation from the 10–30% range would predict negligible degradation at 50%, yet the actual decline is 28%. By 70%, the strategy is essentially destroyed. This pattern is consistent with a phase-transition-like threshold driven by the positive feedback structure encoded in the model (Section 2).

3.2 Market-Level Consequences

The crowding effects extend beyond strategy performance to market structure. Table 2 reports detailed market stability metrics.

Table 2: Market Microstructure Effects of VT Crowding

ϕ	Ann. Vol	VIX Mean	VIX Std	Skewness	MDD	VIX Spike (%)[b]
0%	16.0%	19.4	1.76	−0.005	−33.4%	0.0
10%	16.0%	19.4	1.76	0.001	−34.2%	0.0
20%	16.1%	19.5	1.77	−0.002	−33.2%	0.0
30%	16.6%	20.0	1.89	0.002	−34.9%	0.0
50%	19.0%	22.8	2.33	−0.024	−40.2%	0.3
70%	22.8%	27.4	3.39	−0.321	−60.1%	16.2
100%	35.1%	35.0	6.45	−4.73	−91.3%	90.0

Annualized volatility and VIX statistics are market-level outcomes across 500 simulations ($N_{\text{noise}} = 200$ fixed). MDD is the mean maximum drawdown of the market portfolio.

[b] Percentage of trading days where VIX exceeds the spike threshold (30).

Three patterns emerge. First, market volatility amplification is strongly nonlinear: from near-zero at 30% to +19% at 50% and +119% at 100%. Second, the VIX distribution shifts dramatically—at 70% adoption, VIX spends 16% of days in spike territory, compared to near 0% in the uncrowded baseline, and at 100% this reaches 90%. Third, return skewness turns sharply negative at high adoption levels (−0.32 at 70%, −4.7 at 100%), indicating that VT crowding manufactures left-tail events—precisely the risk that VT is designed to mitigate.

3.3 Statistical Significance

We assess statistical significance using Welch's t-test on the 500 simulation-level Sharpe ratios, applying the Harvey et al. [2016] threshold of $|t| > 3.0$. The 10% versus 30% comparison yields $t = 0.05$ ($p = 0.96$), confirming that these two adoption levels are statistically indistinguishable—consistent with the "safe zone" interpretation below 30%. The degradation at 50% versus 10% yields $t = 7.12$ ($p < 0.001$), far exceeding the Harvey threshold and confirming that the tipping point between 30% and 50% represents a genuine structural break rather than simulation noise.

3.4 The Feedback Structure

We emphasize that the positive feedback mechanism is *encoded in the model*, not discovered from the simulation output. The VIX responds to realized volatility (Eq. 2), VT agents respond to VIX, and their trading affects prices (Eq. 1), which in turn affects realized volatility. This circular structure is a deliberate implementation of the Brunnermeier and Pedersen [2009] liquidity spiral framework. What the simulation *does* reveal is the *quantitative threshold behavior*: the feedback loop's consequences are negligible below 30% adoption, become material at 50%, and are catastrophic at 70%. When VIX rises, the aggregate VT selling pressure is:

$$\text{Sell pressure} \propto \phi \cdot N \cdot \left(\frac{12}{\text{VIX}_{t-1}} - \frac{12}{\text{VIX}_{t-2}} \right), \tag{3}$$

which grows linearly in ϕ but produces *nonlinear* market impact because the selling itself amplifies subsequent VIX increases. The simulation thus serves as a computational laboratory for quantifying a mechanism whose qualitative direction is known, but whose critical mass has not previously been estimated.

3.5 Sensitivity Analysis

To assess whether the 50–70% tipping point is an artifact of our baseline calibration, we vary λ (market impact), γ (VIX sensitivity to realized vol), and κ (VIX mean-reversion speed) each by $\pm 50\%$, producing 27 parameter combinations tested at $\phi \in \{10\%, 30\%, 50\%\}$ with 200 simulations per cell (Table 3).

Table 3: Sensitivity of VT Tipping Point to Key Parameters (Fixed Liquidity)

Parameter	Value	Sharpe_{10}	Sharpe_{30}	Sharpe_{50}	Threshold[c]
	0.0025 (−50%)	0.52	0.49	0.43	>50%
λ	0.005 (baseline)	0.49	0.45	0.35	>50%
	0.0075 (+50%)	0.52	0.40	0.23	30–50%
	100 (−50%)	0.47	0.51	0.36	>50%
γ	200 (baseline)	0.52	0.47	0.32	>50%
	300 (+50%)	0.48	0.41	0.31	>50%
	0.015 (−50%)	0.45	0.48	0.32	>50%
κ	0.03 (baseline)	0.49	0.47	0.34	>50%
	0.045 (+50%)	0.51	0.44	0.32	>50%

Each row varies one parameter while holding others at baseline. 200 simulations per cell ($N_{\text{noise}} = 200$ fixed).

[c] Threshold is the adoption region where Sharpe degradation first exceeds 50% relative to the 10% baseline. Under fixed liquidity, only high-λ produces a threshold below 50%. λ: Kyle market impact; γ: VIX sensitivity to realized volatility; κ: VIX mean-reversion speed.

The tipping point is *conditionally* stable. Under fixed background liquidity, eight of nine parameter combinations produce a threshold above 50%; only the high-λ scenario (+50%) pulls the threshold into the 30–50% region. VIX dynamics (γ, κ) have limited influence: all six γ/κ variants produce a threshold above 50%. Market impact (λ) remains the dominant driver: the Sharpe spread across λ variants at $\phi = 50\%$ is 0.21 (0.43 to 0.23), compared to 0.05 for γ and 0.02 for κ. This is economically intuitive—the feedback loop's severity depends on how much correlated selling moves prices, not on how VIX tracks volatility.

3.6 Design Validation: Fixed vs. Scaled Liquidity

An earlier version of this model scaled noise traders proportionally with $(1-\phi)$, so that at high VT adoption the number of background liquidity providers fell dramatically (from 300 to near zero at $\phi = 100\%$). This confounded two effects: crowding (correlated VT trading) and liquidity evaporation (fewer noise traders to absorb order flow). We address this by fixing $N_{\text{noise}} = 200$ across all adoption levels.

The comparison is revealing. When noise traders are scaled with VT adoption, the threshold appears at 30–50% (Sharpe collapses from 0.43 to 0.18 between 30% and 50%). Fixing background liquidity shifts the threshold to 50–70% (Sharpe falls from 0.47 to 0.34 at 50%, and only collapses to 0.08 at 70%). This demonstrates that approximately half of the originally observed degradation was driven by liquidity evaporation rather than crowding per se. The fixed-liquidity design is more conservative—it attributes less degradation to crowding—and we regard these as the more reliable estimates.

4 Discussion

4.1 Policy Implications

Regulators should monitor VT adoption rates as a systemic risk indicator. The 50–70% tipping point—while far from current levels below 5%—provides a quantitative benchmark. The ESRB and FSOC could incorporate VT adoption metrics into macroprudential surveillance, alongside existing crowding measures [Baltas, 2019].

4.2 Implications for Practitioners

At current adoption levels, VT remains viable. However, the nonlinear tipping point means degradation could accelerate rapidly once critical mass is reached. Practitioners should monitor proxy indicators: the correlation between VIX changes and equity fund flows, volatility-linked AUM concentration, and anomalous intraday reversals during VIX spikes.

4.3 Limitations

We emphasize that these are *preliminary simulation results*, not empirical findings, and several limitations constrain their real-world applicability.

First, our model uses a constant λ, whereas Kyle [1985] derives lambda endogenously from the information structure. In real markets, liquidity evaporates during stress as market makers widen spreads, which would amplify the feedback loop and potentially lower the tipping point. Conversely, in normal times, deep liquidity would moderate price impact. Our sensitivity analysis (Table 3) confirms that λ is the most influential parameter, suggesting that endogenizing it is the highest-priority extension.

Second, agents within each class are homogeneous: all VT agents use identical 12/VIX rules with the same rebalancing frequency. In practice, VT strategies vary widely in their volatility estimators, target levels, rebalancing frequencies, and position limits. This heterogeneity would likely smooth the tipping point and shift it to higher adoption levels.

Third, the model excludes transaction costs, margin constraints, and short-selling restrictions, all of which would moderate extreme outcomes. It also lacks adaptive learning—agents do not switch strategies in response to poor performance, whereas real investors would eventually exit a degraded strategy.

Fourth, our simplified VIX is a realized-volatility tracker that omits the variance risk premium embedded in the true CBOE VIX (option-implied volatility). This distinction may affect the magnitude of the feedback—the real VIX typically exceeds realized volatility, so VT agents in practice hold lower weights than our model implies—but does not alter the direction of the crowding mechanism.

Fifth, with $N = 1{,}000$ agents, the model is small relative to real markets with millions of participants. The price impact per agent is correspondingly larger than in reality.

Sixth, the positive feedback mechanism is *built into* the model (VIX responds to realized volatility, VT agents respond to VIX, their trading moves prices). We do not claim to discover this feedback—it is a structural feature of the simulation—but rather to quantify the adoption level at which it becomes destabilizing.

These limitations suggest that our quantitative thresholds (50–70%) should be interpreted as order-of-magnitude estimates rather than precise critical values. The design validation (Section 3.6) demonstrates that even the *direction* of the threshold depends on modeling choices—an earlier design with scaled liquidity produced a threshold 20 percentage points lower. The emerging empirical literature on VT crowding, including ongoing work on realized VT fund flows, will be essential for calibrating these thresholds to real markets. The qualitative finding—that VT strategies exhibit a nonlinear crowding tipping point whose severity depends on market impact—is more robust than any specific numerical threshold.

5 Conclusion

This paper presents preliminary agent-based evidence that volatility targeting strategies exhibit a nonlinear crowding tipping point. Using a Kyle (1985) market microstructure with 1,000 heterogeneous agents, 200 fixed noise traders, and 500 Monte Carlo simulations, we find that VT performance remains stable below 30% adoption but degrades sharply between 50% and 70%. The model encodes a Brunnermeier–Pedersen positive feedback structure, and we quantify—rather than discover—its threshold behavior under controlled conditions. At 70% adoption, the strategy is essentially destroyed (Sharpe 0.08) and market kurtosis rises to 1.4; at 100%, kurtosis reaches 61. A design validation confirms that approximately half the degradation in a naïve specification is attributable to liquidity evaporation rather than crowding. Sensitivity analysis across 27 parameter combinations confirms that the threshold is conditionally stable—shifting primarily with market impact (λ) while remaining robust to VIX dynamics.

Current VT adoption appears safely below this threshold. However, the nonlinear nature of the tipping point—where small increases in adoption produce disproportionately large degradation—suggests that monitoring VT growth is warranted. Future work should extend the model with endogenous λ, heterogeneous VT specifications, adaptive strategy switching, and calibration to real market microstructure data to refine the threshold estimate.

References

Baltas, N. (2019). The impact of crowding in alternative risk premia investing. *Financial Analysts Journal*, 75(3), 89–104.

Bookstaber, R., Paddrik, M., and Tivnan, B. (2014). An agent-based model for financial vulnerability. OFR Working Paper 14-05, Office of Financial Research.

Bouchaud, J.-P., Bonart, J., Donier, J., and Gould, M. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brunnermeier, M. K. and Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.

European Central Bank. (2020). Procyclicality of volatility targeting strategies. *Financial Stability Review*, May.

Gennotte, G. and Leland, H. E. (1990). Market liquidity, hedging, and crashes. *American Economic Review*, 80(5), 999–1021.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). …and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M., and Van Hemert, O. (2018). The impact of volatility targeting. *Journal of Portfolio Management*, 45(1), 14–33.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.

Moreira, A. and Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.

Perchet, R., de Carvalho, R. L., Heckel, T., and Moulin, P. (2015). Predicting the success of volatility targeting strategies: Application to equities and other asset classes. *Journal of Alternative Investments*, 18(3), 21–38.

Cole, C. (2017). Volatility and the alchemy of risk. Artemis Capital Management, Research Report.